

CS 131 — Midterm 1 Power Sheet

Linux CLI · regex/grep · redirection · permissions · processes · env · git + worked ws1–ws3 solutions, lookup tables & a goal→command index

REMOTE · GIT BASICS (slide 1)

ssh user@host log in to remote Linux (DGX) server

multi-user box — permissions protect users from each other

git: clone status add commit push (repo must be public)

PATHS · ls · WILDCARDS

. cwd . .parent → home /root · abs starts at /
pwd cd . . ls file(type) cat/less man cmd
ls -llong -ahidden -hhuman -rrev -mtime
-Ssize
cp -r mv old new mkdir -p rm -ri(no undo)
ln -s

*any ?one [abc]one [^abc]not
[0-9]/[a-z]range

Data?.csv#Data10 · a[0-9][0-9].txt→a05

REGEX TOKENS · grep -E

. any1 *0+ +1+ ?0/1 . ^start \$end
[abc]one [^b]not [a-z]range
[:digit:]class
{n}exactly {n,m}range a|bOR ()group
\escape: \.dot \\$\\$ *star · class needs [[:]]
BRE=grep(+?{}|())literal → prefer -E (ERE)
POSIX: [[:digit:]] [[:alpha:]]
[[:upper:]] [[:space:]]

REGEX EXAMPLES

.at->hat,cat · ^[hc]at\$→EXACTLY hat/cat
[^b]at->any _at but bat ·
a[[:digit:]]b->a0b–a9b
[0-9]{3}-[0-9]{3}-[0-9]{4}->phone
NNN-NNN-NNNN
"stars":(4\.[5|5\.[0-9])>4.5/5.0 (escape the dot!)
^([^\,]+,222,->222 in 2nd CSV col
\w+@\w+.\w+>email · ^[A-Z]+>leading CAPS

grep FLAGS

-i case -c count -n line# -v invert -w whole-word
-o only-match -oE extract field -E extended
-r recursive
-l names only -A/-B/-C ncontext lines
-f patterns-from-file

ws1B — grep / Yelp SOLUTIONS

city: grep "city":"Santa Barbara" biz.json
count CA: grep -c "state":"CA" biz.json
worth(any case): grep -i worth review.json
coffee: grep -E 'Coffee|Cafe|Café' biz.json
stars 4.5/5.0: grep -E
"stars":(4\.[5|5\.[0-9]) biz
exactly 5-star: grep -E "stars":5, review.json
emails: grep -E '\w+@\w+.\w+' review.json
phones: grep -E
'[0-9]{3}-[0-9]{3}-[0-9]{4}' f
name P...: grep "name":"P" biz.json
count pizza: grep -ic pizza review.json
service+line#: grep -n service review.json
not Restaurants: grep -v Restaurants biz.json
SB names A→Z: grep "city":"Santa Barbara" biz | grep -oE
"name":"[A-Z]+" | sort
CA review_count top3:
grep "state":"CA" biz | grep -oE
"review_count":[0-9]+" | grep -oE
'[0-9]+' | sort -nr | head -3

REDIRECTION · FD · PIPES

FDs 0stdin 1stdout 2stderr
>overwrite >>append 2>err 2>>err-app &>both
<stdin

2>/dev/null drop err · | cmd->cmd (vs >
cmd->file)

ws3A — I/O & PIPE SOLUTIONS

ls -l > listing.txt; date >>
listing.txt
err only: ls /bad 2> errors.txt
split: ls /bad /etc > out.txt 2> err.txt
both: ls /bad /etc &> combined.txt
silence: ls /bad /etc 2>/dev/null
unique fruits: sort fruits.txt | uniq
counts desc: sort fruits.txt | uniq -c |
sort -nr
count ERROR: grep -c ERROR server.log
ERROR>file: grep ERROR server.log >
errors_only.txt
log levels: grep -oE '^[A-Z]+' server.log
| sort | uniq -c | sort -nr
top5 states: ... | sort | uniq -c | sort -nr
| head -5

TEXT TOOLS

cat join tac reverse nl number sort -n/-r uniq
-c
wc -l/-w/-c head/tail -n N cut -d, -f2
tr a becho

QUOTING & EXPANSION

". ."keep \$ \ . "all literal · \escape 1
\$((2+3))=5 · {1..5}>1 2 3 4 5 ·
file{a,b}>filea fileb
\$HOME param · \$(date) / backticks = command
substitution

vim

i insert a after o new line · Esc=normal mode
dd cut yy copy pp paste /text search nn next u undo
:w save :q quit :wq save+quit :q! quit no-save

ws1A — BASICS SOLUTIONS

abs path back: cd ~/cs131/ws1 then pwd
sibling from logs: ls ../data
file type: file logs/app01.log
all hidden long human rev: ls -alhr
logs by size desc: ls -ls logs/
Data + 1 char: ls data/Data?.csv
uppercase-start: ls [[:upper:]]* (or [A-Z]*)
BACKUP.3 digits: ls
logs/BACKUP.[0-9][0-9][0-9]
copy dir+contents: cp -r data backup
rename: mv data/readme.txt
data/README.md
safe delete: rm -ri backup/data
commit: git add answers.txt && git
commit -m "msg"
clone: git clone <url>

PERMISSIONS

-rwxr-xr-- = type | owner | group | other
OCTAL 0---1--x2-w-3-wx4r--5r-x6rw-7rwx
ex: rwxr-x---=750 · rw-r--r--=644 ·
rw-----=600
chmod 640 f(octal) · symbolic u g o a + - =
chmod u+x f chmod go-w f · only owner/root
may chmod
umask 022->files 644 dirs 755 · id=uid/gid/groups
chgrp grp f chown me f(root) su -(target pw)
sudo(your pw)

ws3B — PERMS / find SOLUTIONS

owner rw/group r/other none: chmod 640
fruits.txt
strip→restore: chmod 000 f then chmod u+rw f
my writable files: find ws3 -type f -user
\$USER -perm -u+w

PROCESSES · JOBS

ps(-p PID,aux) top live jobsthis shell's jobs
cmd & bg fg %1fore Ctrl-Z suspend bg
%1 resume
disown detach(survives close) · init=PID 1(first
proc)

SIGNALS

1 SIGHUP (terminal closed) · 2 SIGINT (Ctrl-C)
9 SIGKILL (force) · 15 SIGTERM (kill default) ·
TSTP (Ctrl-Z)

ws3C — JOBS SOLUTIONS

bg job: bash long.sh > log.txt &
suspend→bg: Ctrl-Z then bg %1 · fore: fg %1
kill job: kill %1 · detach: ... & then disown %1

ENV · STARTUP

printenv export V=x alias ll='ls -l'
login→/etc/profile, ~/.bash_profile ·
non-login→ ~/.bashrc
hidden=.name(ls -a) · reload source
~/.bashrc

cron · find · diff · curl

cron min hour dom mon dow cmd · *=every ·
crontab -e
30 08 * * * cmd=8:30 daily · find P -name
'*.x' -size +1M
find size c/k/M/G · diff a b · curl -o f URL

GIT ESSENTIALS

clone status add commit -m push pull
log --oneline -3
fetch=download only (pull=fetch+merge)
switch -c name merge name diff a b
branch -d name

ws2 — GIT SOLUTIONS

commit: git add notes.md && git commit
-m "msg"
last 3: git log --oneline -3
undo pushed: git revert <hash> (keeps
history)
big file: git rm --cached big.bin
+ .gitignore, git commit --amend, git push
(>100MB)
branch: git switch -c feature-summary
diff branches: git diff main
feature-summary
merge: git switch main && git merge
feature-summary
delete merged: git branch -d
feature-summary
conflict: edit <<<<<< ===== >>>>>> → add
→ commit
fetch+incoming: git fetch origin
git log HEAD..origin/main --oneline
GOAL → COMMAND (quick index)
count matches → grep -c · only the match →
grep -o
rank duplicates → sort | uniq -c | sort
-nr
2nd CSV column → cut -d, -f2 or ^([^\,]+,X,
both streams→file → &> · drop errors →
2>/dev/null
keep running after logout → disown (or nohup)
exactly N digits → [0-9]{N} · literal dot → \.
new+switch branch → git switch -c · safe
undo → git revert
dir copy → cp -r · default perms → umask

LOOKUP — ls -l FIELDS

-rw-r--r-- 1 me staff 4096 Jun24 09:00
file
type(-file ddir llink) · perms(u/g/o) · links · owner ·
group · size · date · name

LOOKUP — find TESTS

-name '*.log' -type f | d -size +1M-100k
-user me -perm -u+w -mtime -1 -exec cmd
{ } \;

LOOKUP — cron FIELDS

min0-59 hour0-23 dom1-31 mon1-12 dow0-7(Sun)

LOOKUP — SIGNAL NUMBERS

1 HUP · 2 INT · 9 KILL · 15 TERM · 18 CONT ·
19/20 STOP/TSTP

COMMON GOTCHAS

uniq only drops ADJACENT dups → sort first
escape . in regex (\.) or it matches ANY char
[:digit:] must be [[:digit:]] (double
brackets)
cp a directory needs -r · > OVERWRITES (use
>>)
'..' blocks \$/globs; '..' keeps \$ \ .

only owner/root can **chmod** · JSON field order varies

grep -E for **+ ? { } | ()** (plain grep treats them literal)

EXAM TIPS

Read each **rm/reset** BEFORE Enter — no undo.

Each line = 1 JSON record → anchor on **"field":**.

Manual given for full options — know the flags you USED.

Vague Q? leave a short note (scratch = partial credit).

Bring Student ID · one 2-sided cheat sheet allowed.

CONCEPTS — define these (oral/short Q)

process = running instance of a program (has a PID)

daemon = background service process (e.g. started by init)

init = first process, PID 1, runs until shutdown

signal = async message to a process (INT, KILL, TERM, HUP)

SIGHUP = sent when terminal closes → kills its jobs (**disown** survives)

pipe **|** = stdout of one command → stdin of the next

redirection = change where stdin/stdout/stderr go (files)

file descriptor = process's numbered I/O channel (0/1/2)

/dev/null = bit bucket — discards anything written to it

login shell = prompted user+pw (SSH/console) → **~/.bash_profile**

non-login shell = GUI terminal → **~/.bashrc**

absolute path = from **/** · relative = from cwd

hard link vs symlink = **ln** vs **ln -s** (pointer to a path)

wildcard = shell glob on filenames · regex = grep on text

BRE vs ERE = **grep** vs **grep -E** (operators vs literal)

foreground vs background = blocks terminal vs **&/bg**

env var (exported→children) vs shell var (local)

root/superuser = uid 0 · **/etc/shadow** = hashed pw (root-only)

umask = permission bits REMOVED from the default

PIPELINE COOKBOOK

count lines: **wc -l f** or **... | wc -l**

unique (sorted): **sort f | uniq**

frequency table: **sort f | uniq -c | sort -nr**

nth field: **cut -d, -f2 f**

top N: **... | head -N** · last N: **... | tail -N**

extract+tally: **grep -oE PAT f | sort | uniq -c | sort -nr**

MORE OPTIONS

ls -lone/line -Rrecursive -ddir-itself -iinode

sort -k2field2 -t,delim -uunique -hhuman

find -maxdepth N -iname -empty -newer f

grep -xwhole-line -qquiet -Hshow filename -rrrecurse

chmod -R recursive · cp -iprompt -ppreserve -vverbose

SSH · SCP

ssh user@host log in · **ssh -i key user@host**

scp file user@host:~/path copy to remote

ssh-keygen make keypair · **~/.ssh/config** saved hosts

MORE GIT

git reset --soft/--hard HEAD~1 (REWRITES history — careful)

git stash / git stash pop shelve work

git rebase main replay commits onto main

git remote -v · git push -u origin main set upstream

git checkout -- f discard file changes · **git diff --staged**

EXPANSION — try it

echo {A,B}{1,2} → A1 A2 B1 B2

echo \$((10/3)) → 3 (integer) · **echo \$((2*8))** → 256

echo "\$USER" → name · **echo '\$USER'** → \$USER (literal)

mkdir -p proj/{src,bin,doc} makes 3 subdirs at once

MORE WORKED ITEMS

read **ls -l**: cols =

perms,links,owner,group,size,date,name

id info: **id** → uid, gid, groups (uid=user, gid=group)

predict umask: default 666 – umask = new file mode

short hash of bad commit: **git log --oneline** (copy 7-char id)

count CA businesses returns a number: add **-c** to grep

phone in text: **grep -E**

'[0-9]{3}-[0-9]{3}-[0-9]{4}'
review.json